RadioSkyMap

HI & 408 MHz Synchrotron Visualizer

USER GUIDE

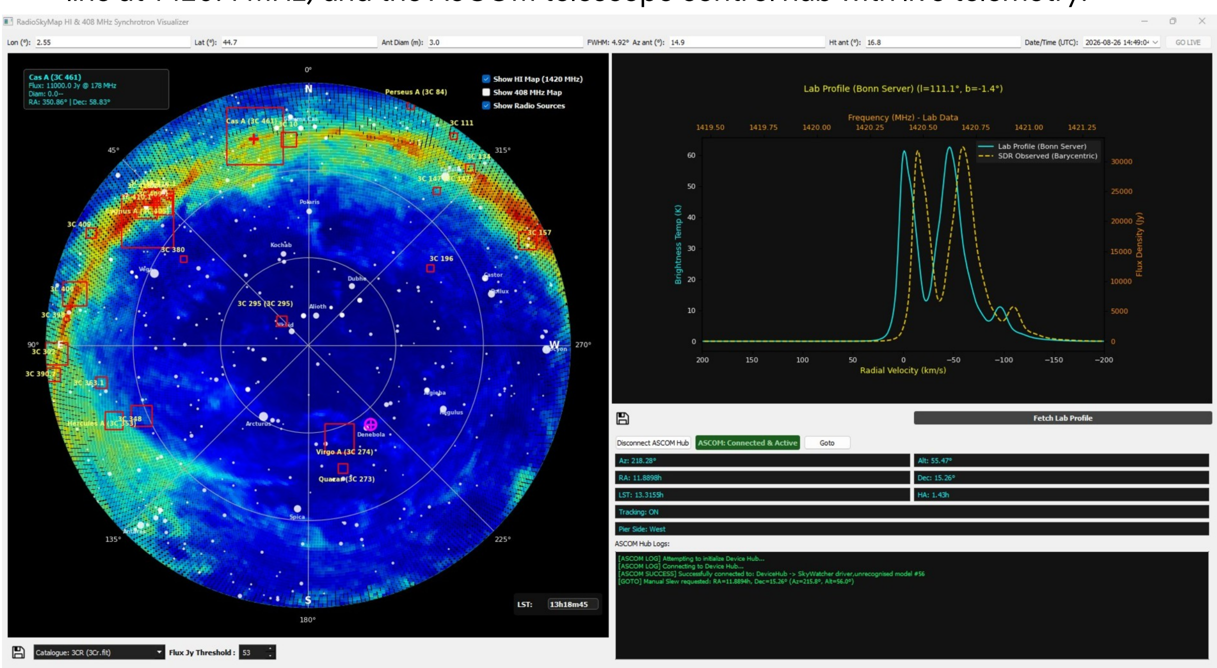
1. Overview

RadioSkyMap is an desktop application designed for radio astronomy enthusiasts and amateur astronomers. It provides a **real-time sky map** with (azimuth/elevation) overlaid with **Neutral Hydrogen (HI) emission data, 408 MHz synchrotron continuum maps**, and major celestial radio source catalogs (such as **3CR, Parkes**, and **G4Jy**). It also integrates seamlessly with **ASCOM-compliant telescope** mounts for automated tracking and pointing ("**GOTO**"), alongside spectral profile data acquisition from the **Bonn HI survey** or local databases.

2. Main Interface Layout

The user interface is split into three main interactive sections:

- **Top Control Bar:** Configures your geographic coordinates (Longitude, Latitude), antenna parameters (Diameter, Frequency, Azimuth, Elevation), UTC date/time, and live synchronization mode.
- **Polar Sky Map (Left Panel):** Displays a real-time 360° horizon view with cardinal points (N, E, S, W), bright stars, solar system bodies (Sun, Moon, Jupiter), and radio overlays.
- **Spectra & ASCOM Panel (Right Panel):** Houses the spectral velocity profile graph (HI line at 1420.4 MHz) and the ASCOM telescope control hub with live telemetry.



3. Key Features & Operations

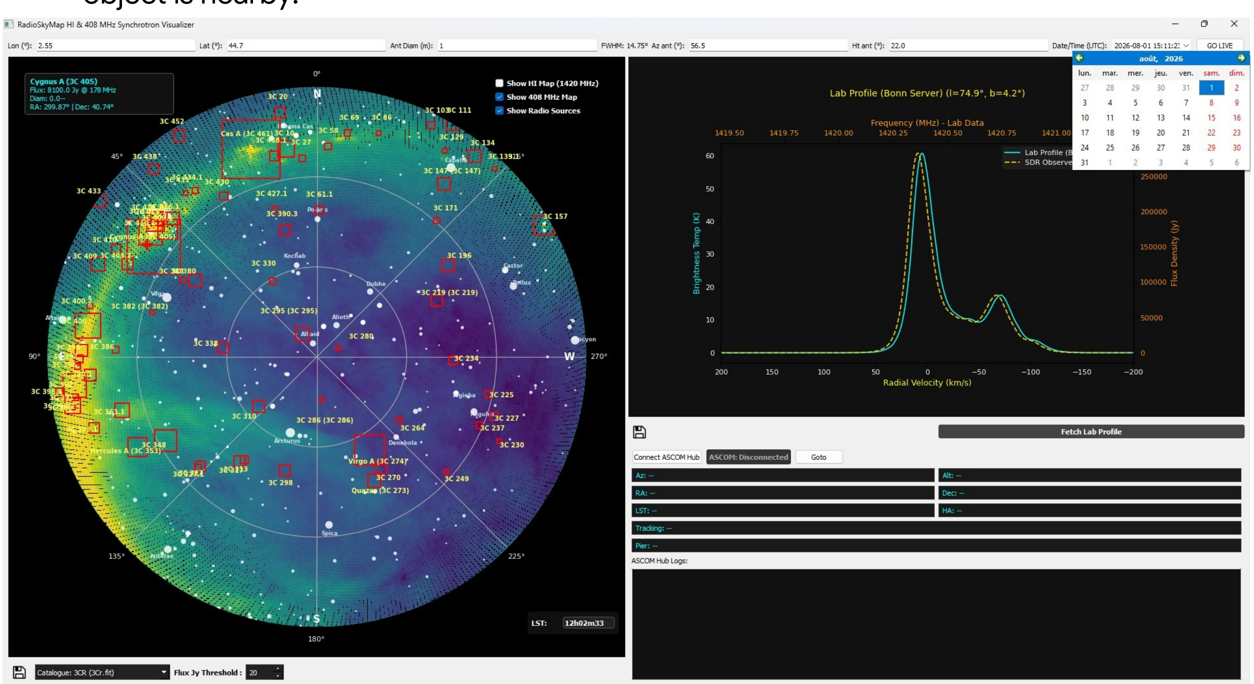
A. Geographic & Antenna Configuration

- **Location Settings:** Enter your station's **Longitude** and **Latitude** in decimal degrees. These values are automatically saved in config.ini upon editing.
- **Beamwidth Calculation (FWHM):** Based on your antenna diameter (in meters) and observation frequency (in MHz), the application automatically computes and displays the Half-Power Beamwidth using the standard formula:
  FWHM = 1.22 * (λ/D)
- **Manual vs. Live Time Mode:** By default, the application runs in **Live Mode**, synchronizing with your computer's current UTC time and updating the Local Sidereal Time (LST). Changing the date/time manually via the calendar picker switches the app to Custom Mode, which can be reverted by clicking **GO LIVE**.

B. Polar Sky Map & Overlays

- **Interactive Layers:** Using the overlay checkboxes on the map, you can toggle:
  - *HI Map (1420 MHz):* Neutral hydrogen gas distribution from survey data.
  - *408 MHz Map:* Synchrotron continuum background radiation.
  - *Radio Sources:* High-flux cosmic sources (e.g., Cygnus A, Cassiopeia A, Taurus A) filtered dynamically by a user-defined flux threshold (in Jansky / Jy).

    * Slowdowns prior to display are sometimes observed, due to data calculations and loading.
- **Mouse Interaction:** Clicking anywhere on the polar map instantly updates the antenna target coordinates (Azimuth/Elevation) and queries database information if a celestial object is nearby.



C. Spectral Profile Fetching ("Fetch Lab Profile")

- By clicking the **Fetch Lab Profile** button, the app queries the Bonn HI survey server or falls back to a local SQLite database (lab_sky_survey.db).
- It plots the brightness temperature (T_b) against radial velocity (km/s) and provides secondary axes for frequency and flux density (Jy).

D. ASCOM Telescope Integration

- If you have a mount controllable via the ASCOM standard, and you have already configured and successfully tested the **ASCOM Hub**, you can use this software feature to control your mount via RadioSkyMap.
  You must first set up your mount and its electronics.
- **Connection:** Click "**Connect ASCOM Hub**" to link the software to your ASCOM-compatible telescope driver/mount.
- **Telemetry:** Real-time data streams include azimuth, altitude, right ascension (RA), declination (Dec), LST, hour angle (HA), tracking status, and pier side.
- **GOTO Pointing:** Click on an object on the map and use the "**Goto**" button to automatically send the target coordinates to your telescope mount.
  The telescope mount will automatically point to the object.